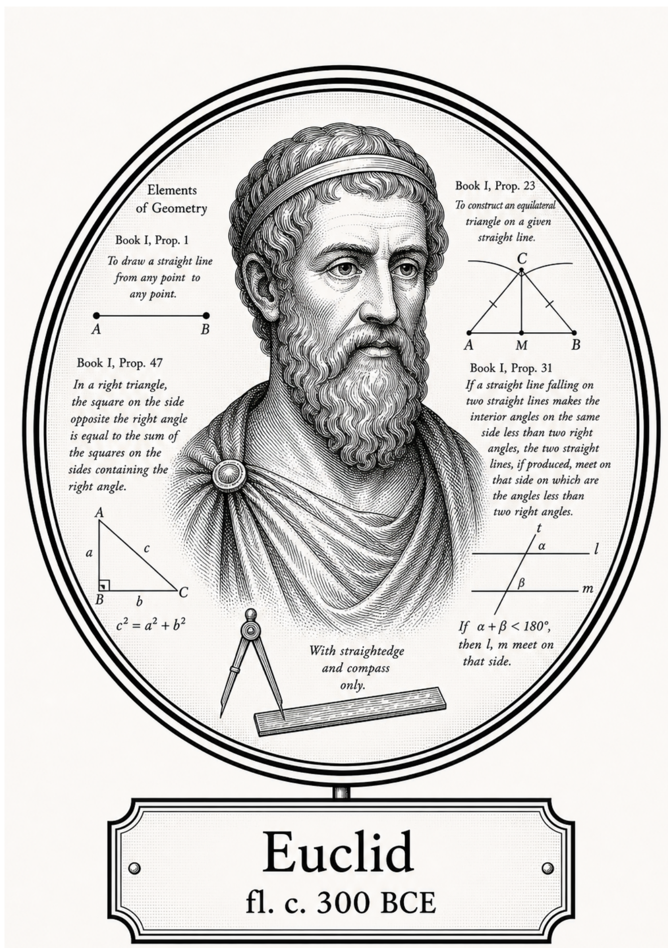
From Cantor to Ito

Logic, Sets, and Proof

Foundational Geometry



Euclid

fl. c. 300 BCE

Self-Study Notes

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July 2, 2026

For every reader learning to make mathematics their own.

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Euclidean Geometry

1.1 Foundations

Definitions

Remark (Source note). The following definitions are Euclid's Book I definitions in the Heath translation [Euclid, 1956, Book I, Definitions 1–23].

Definition

Definition 1.1.1 (Point)**.** A point is that which has no part.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.2 (Line)**.** A line is breadthless length.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.3 (Ends of a line)**.** The ends of a line are points.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.4 (Straight line)**.** A straight line is a line which lies evenly with the points on itself.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.5 (Surface)**.** A surface is that which has length and breadth only.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.6 (Edges of a surface)**.** The edges of a surface are lines.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.7 (Plane surface)**.** A plane surface is a surface which lies evenly with the straight lines on itself.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.8 (Plane angle)**.** A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.9 (Rectilinear angle)**.** And when the lines containing the angle are straight, the angle is called rectilinear.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.10 (Right angle and perpendicular)**.** When a straight line standing on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.11 (Obtuse angle)**.** An obtuse angle is an angle greater than a right angle.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.12 (Acute angle)**.** An acute angle is an angle less than a right angle.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.13 (Boundary)**.** A boundary is that which is an extremity of anything.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.14 (Figure)**.** A figure is that which is contained by any boundary or boundaries.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.15 (Circle)**.** A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure equal one another.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.16 (Center of a circle)**.** And the point is called the center of the circle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.17 (Diameter)**.** A diameter of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.18 (Semicircle)**.** A semicircle is the figure contained by the diameter and the circumference cut off by it. And the center of the semicircle is the same as that of the circle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.19 (Rectilinear figures)**.** Rectilinear figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.20 (Equilateral, isosceles, and scalene triangles)**.** Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.21 (Right-angled, obtuse-angled, and acute-angled triangles)**.** Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acute-angled triangle that which has its three angles acute.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.22 (Quadrilateral classes)**.** Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.23 (Parallel straight lines)**.** Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Postulates

Remark (Source note). The following are Euclid's five postulates from Book I of the *Elements* in the Heath translation [Euclid, 1956, Book I, Postulates 1–5].

Remark (Exposition). Postulates are geometry-specific starting assumptions. They say which constructions are permitted and which basic geometric facts may be used without proof. In Euclid's system, these are not conclusions reached from earlier results; they are part of the foundation from which the theory begins.

Later propositions are proved from the definitions, the postulates, the common notions, and propositions already established. Thus the postulates are the rules of play for Euclidean geometry: they determine what can be constructed, what may be assumed about straight lines and circles, and where the theory's geometric content enters before proof begins.

Axiom

Axiom 1.1.24 (Euclid Postulate I: Drawing a straight line)**.** Let it have been postulated to draw a straight line from any point to any point.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.25 (Euclid Postulate II: Producing a finite straight line)**.** Let it have been postulated to produce a finite straight line continuously in a straight line.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.26 (Euclid Postulate III: Describing a circle)**.** Let it have been postulated to describe a circle with any center and distance.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.27 (Euclid Postulate IV: Equality of right angles)**.** Let it have been postulated that all right angles are equal to one another.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.28 (Euclid Postulate V: Parallel postulate)**.** Let it have been postulated that, if a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axioms

Remark (Source note)*.* Euclid calls the following principles *common notions*. They function as general axioms governing equality and comparison in Book I of the *Elements* [Euclid, 1956, Book I, Common Notions 1–5].

Axiom

Axiom 1.1.29 (Euclid Common Notion I: Equality is transitive through a common equal)**.** Things which are equal to the same thing are also equal to one another.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.30 (Euclid Common Notion II: Adding equals to equals)**.** If equals be added to equals, the wholes are equal.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.31 (Euclid Common Notion III: Subtracting equals from equals)**.** If equals be subtracted from equals, the remainders are equal.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.32 (Euclid Common Notion IV: Coincidence implies equality)**.** Things which coincide with one another are equal to one another.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.33 (Euclid Common Notion V: Whole and part)**.** The whole is greater than the part.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Remark (Exposition). Common notions are general principles of reasoning rather than specifically geometric assumptions. They express basic behavior of equality, comparison, and logical transfer: equal things may be substituted for equal things, adding or subtracting equals preserves equality, and a whole exceeds its part.

Because these principles are not tied to points, lines, circles, or planes, they can be used throughout mathematics, not only in geometry. In modern language, they play the role of background axioms governing how equality and comparison may be used inside proofs.

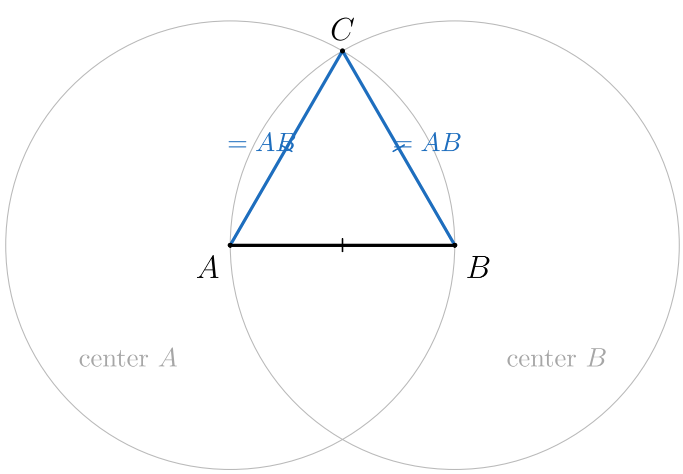
1.2 Compass-and-Straightedge Constructions

Propositions

Remark (Source note). The following propositions are selected from Euclid's Book I in modern theorem form, following the Heath translation of the *Elements* [Euclid, 1956, Book I]. The selection is curated for the trigonometric construction: congruence, parallel-angle control, triangle angle-sum, area comparison, square construction, and the Pythagorean theorem.

Theorem

Theorem 1.2.1 (Euclid I.1: Constructing an equilateral triangle)**.** *Given a finite straight line segment AB, there exists an equilateral triangle ABC constructed on AB.*

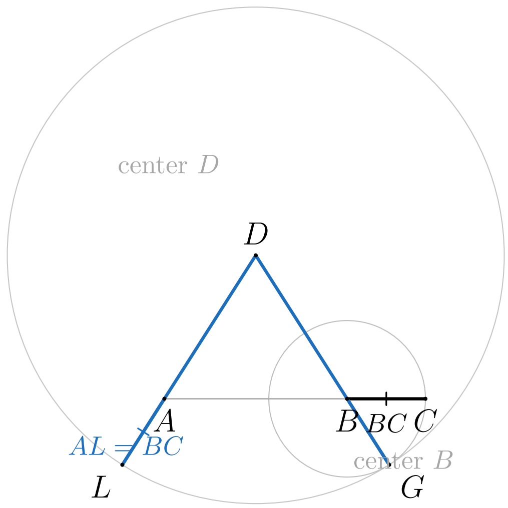


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.2 (Euclid I.2: Copying a segment from a given point)**.** *Given a point A and a finite straight line segment BC, there exists a straight line segment beginning at A and equal to BC.*

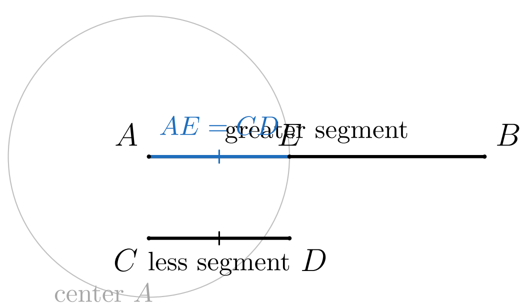


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.3 (Euclid I.3: Cutting off an equal segment)**.** *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

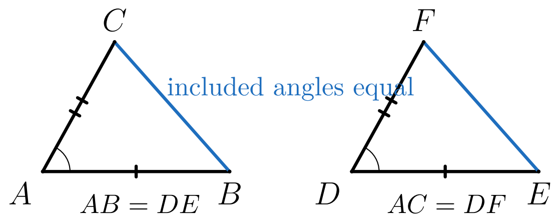


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.4 (Euclid I.4: Side-angle-side congruence)**.** *If two triangles have two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, then they also have the base equal to the base, the triangle equal to the triangle, and the remaining angles equal to the remaining angles respectively.*

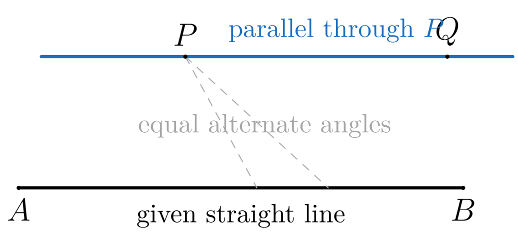


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.5 (Euclid I.31: Drawing a parallel through a point)**.** *Through a given point, to draw a straight line parallel to a given straight line.*

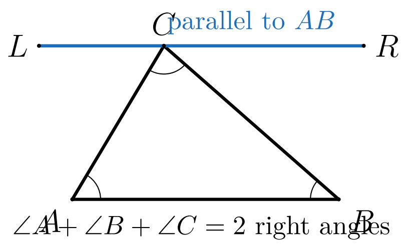


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.6 (Euclid I.32: Triangle angle sum)**.** *In any triangle, if one of the sides is produced, then the exterior angle is equal to the two interior and opposite angles, and the three interior angles of the triangle are equal to two right angles.*

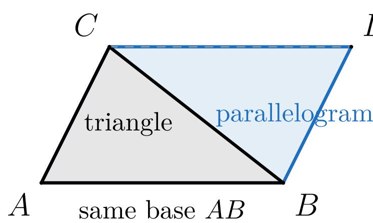


Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.7 (Euclid I.41: Triangle and parallelogram on the same base)**.** *If a parallelogram has the same base with a triangle and is in the same parallels, then the parallelogram is double the triangle.*

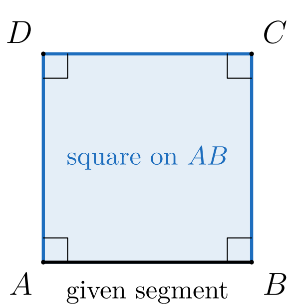


Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.8 (Euclid I.46: Constructing a square on a segment)**.** *On a given straight line, to describe a square.*

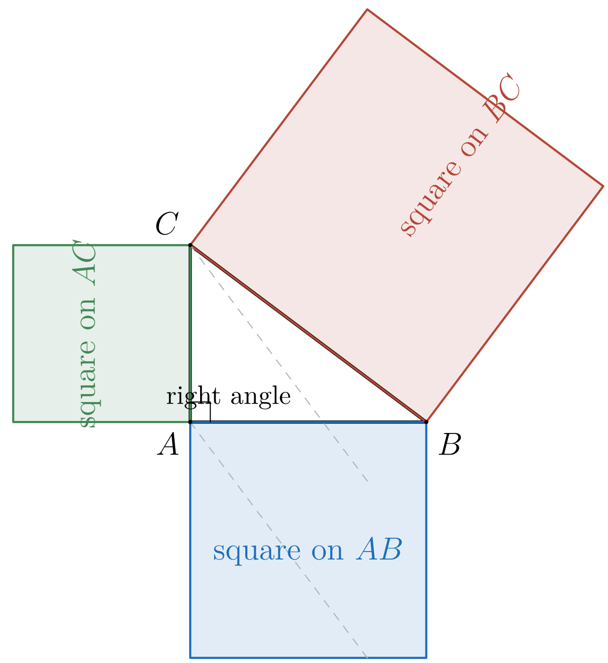


Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.9 (Euclid I.47: Pythagorean theorem)**.** *In right-angled triangles, the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.*



Remark (Proof). Go to proof.

Remark (Placement). This proposition is pulled forward as the Book I capstone needed by the trigonometric definitions. Its professional proof depends on later Book I propositions that are not yet developed in this chapter; the proof file records those pending Euclidean nodes explicitly.

Chapter 2

Trigonometry

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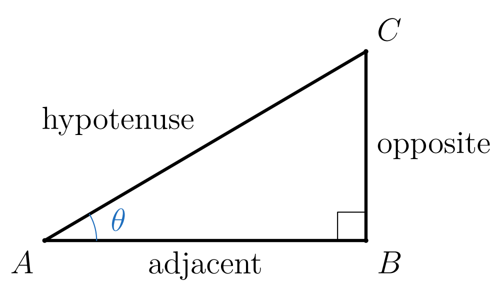
VOLUME I
Logic, Sets, and Proof

BOOK III
Foundational Geometry

CHAPTER 11
Trigonometry

2.0.1 Trigonometric Functions from Euclidean Geometry

Remark (Geometric intuition). A right triangle fixes three side lengths relative to an acute angle: opposite, adjacent, and hypotenuse. Similar right triangles may be larger or smaller, but their corresponding side ratios are unchanged. This invariance is what makes a trigonometric ratio a function of the angle rather than a feature of one drawing.



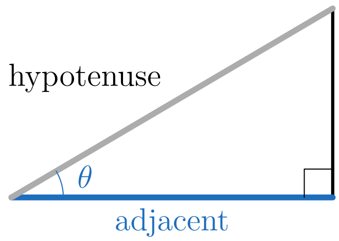
Definition

Definition 2.0.1 (Right-triangle sine)**.** Let T be a right triangle and let θ be one of its acute angles. Write

$$\mathrm{opp}(T, \theta), \qquad \mathrm{adj}(T, \theta), \qquad \mathrm{hyp}(T)$$

for the side lengths opposite θ, adjacent to θ, and opposite the right angle, respectively. Define

$$\sin \theta = \frac{\mathrm{opp}(T, \theta)}{\mathrm{hyp}(T)}.$$

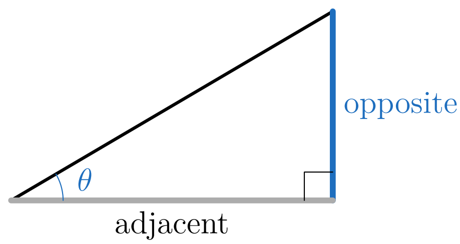


$\cos\theta = \text{adjacent}/\text{hypotenuse}$

Definition

Definition 2.0.2 (Right-triangle cosine)**.** Let T be a right triangle and let θ be one of its acute angles. Define

$$\cos\theta = \frac{\operatorname{adj}(T,\theta)}{\operatorname{hyp}(T)}.$$

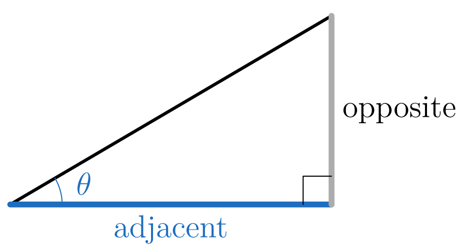


$\tan\theta = \text{opposite}/\text{adjacent}$

Definition

Definition 2.0.3 (Right-triangle tangent)**.** Let T be a right triangle and let θ be one of its acute angles. Define

$$\tan\theta = \frac{\operatorname{opp}(T,\theta)}{\operatorname{adj}(T,\theta)}.$$

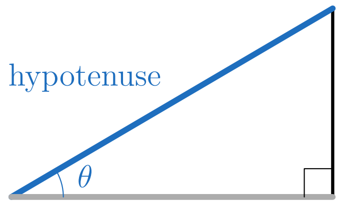


$\cot\theta = \text{adjacent}/\text{opposite}$

Definition

Definition 2.0.4 (Right-triangle cotangent)**.** Let T be a right triangle and let θ be one of its acute angles. Define

$$\cot\theta = \frac{\operatorname{adj}(T,\theta)}{\operatorname{opp}(T,\theta)}.$$

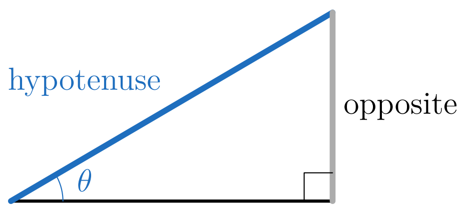


$\sec\theta = \text{hypotenuse}/\text{adjacent}$

Definition

Definition 2.0.5 (Right-triangle secant)**.** Let T be a right triangle and let θ be one of its acute angles. Define

$$\sec\theta = \frac{\mathrm{hyp}(T)}{\mathrm{adj}(T,\theta)}.$$



$\csc\theta = \text{hypotenuse}/\text{opposite}$

Definition

Definition 2.0.6 (Right-triangle cosecant)**.** Let T be a right triangle and let θ be one of its acute angles. Define

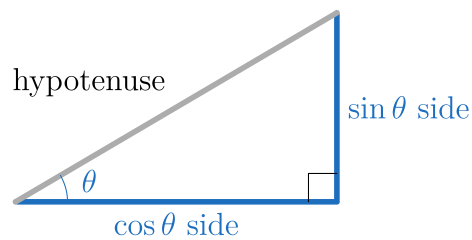
$$\csc\theta = \frac{\mathrm{hyp}(T)}{\mathrm{opp}(T,\theta)}.$$

Remark (Standard quantified statement). For every right triangle T and every acute angle θ of T, the six displayed ratios are defined whenever the denominator in the displayed quotient is nonzero.

Remark (Primitive and derived ratios). The primitive geometric ratios are sine and cosine. Tangent, cotangent, secant, and cosecant are convenient quotient or reciprocal ratios built from the same three side lengths.

Remark (Dependencies).

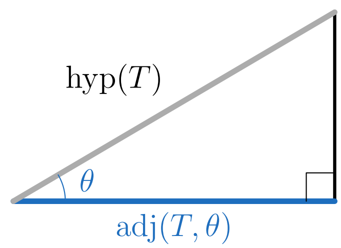
- Right-angled triangle
- Right angle and perpendicular



Definition

Definition 2.0.7 (Acute-angle sine)**.** For an acute Euclidean angle θ, choose any right triangle T having θ as an acute angle and set

$$\sin\theta = \frac{\operatorname{opp}(T,\theta)}{\operatorname{hyp}(T)}.$$



$\cos\theta = \operatorname{adj}(T,\theta)/\operatorname{hyp}(T)$

Definition

Definition 2.0.8 (Acute-angle cosine)**.** For an acute Euclidean angle θ, choose any right triangle T having θ as an acute angle and set

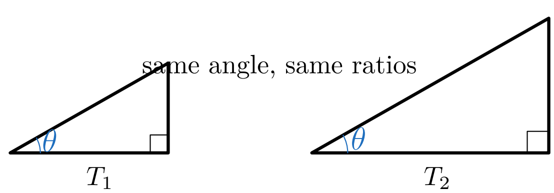
$$\cos\theta = \frac{\operatorname{adj}(T,\theta)}{\operatorname{hyp}(T)}.$$

Remark (Standard quantified statement)*.* For every acute Euclidean angle θ, if T is a right triangle having θ as an acute angle, then $\sin\theta$ and $\cos\theta$ are the opposite-over-hypotenuse and adjacent-over-hypotenuse ratios of T.

Remark (Well-definedness debt)*.* These definitions depend on the similarity-invariance theorem below. Until that theorem is invoked, the displayed formulas describe a ratio in a chosen triangle, not yet a function of θ alone.

Remark (Dependencies)*.*

- Right-triangle sine
- Right-triangle cosine



Theorem

Theorem 2.0.9 (Similarity invariance of trigonometric ratios)**.** *If two right triangles have the same acute angle θ, then their corresponding opposite-to-hypotenuse and adjacent-to-hypotenuse ratios are equal.*

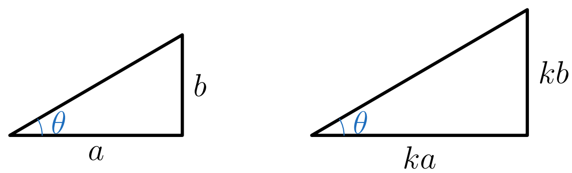
Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.* For all right triangles T_1, T_2 and every acute angle θ, if θ is an acute angle of both triangles, then

$$\frac{\mathrm{opp}(T_1,\theta)}{\mathrm{hyp}(T_1)} = \frac{\mathrm{opp}(T_2,\theta)}{\mathrm{hyp}(T_2)} \quad \text{and} \quad \frac{\mathrm{adj}(T_1,\theta)}{\mathrm{hyp}(T_1)} = \frac{\mathrm{adj}(T_2,\theta)}{\mathrm{hyp}(T_2)}.$$

Remark (Dependencies)*.*

- Euclid I.32



same angle, same ratios

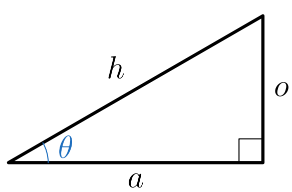
Theorem

Theorem 2.0.10 (Well-definedness of sine and cosine)**.** *For each acute Euclidean angle θ, the values $\sin\theta$ and $\cos\theta$ are independent of the right triangle chosen to compute them.*

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Acute-angle sine
- Acute-angle cosine
- Similarity invariance of trigonometric ratios



$$\begin{aligned} \sin &= o/h, & \cos &= a/h \\ \tan &= o/a, & \cot &= a/o \\ \sec &= h/a, & \csc &= h/o \end{aligned}$$

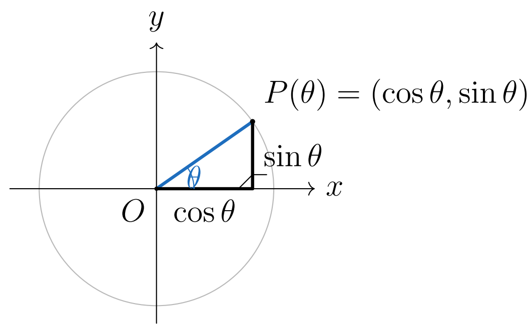
Theorem

Theorem 2.0.11 (Six trigonometric functions)**.** *The six right-triangle trigonometric ratios define functions of an acute Euclidean angle, subject to the denominator restrictions in their defining quotients.*

Remark (Proof). Go to proof.

Remark (Dependencies).

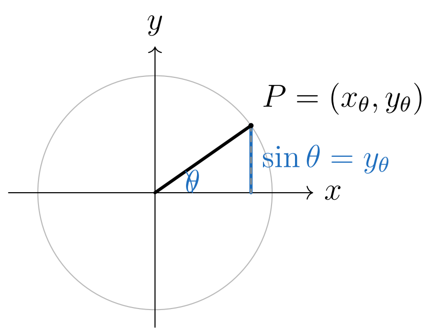
- Right-triangle sine
- Right-triangle cosine
- Well-definedness of sine and cosine



Definition

Definition 2.0.12 (Unit-circle cosine)**.** Let θ be an angle in standard position, and let $P(\theta) = (x_\theta, y_\theta)$ be the point where its terminal ray meets the unit circle. Define

$$\cos\theta = x_\theta.$$



Definition

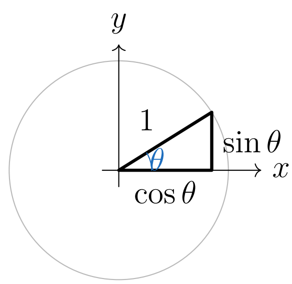
Definition 2.0.13 (Unit-circle sine)**.** Let θ be an angle in standard position, and let $P(\theta) = (x_\theta, y_\theta)$ be the point where its terminal ray meets the unit circle. Define

$$\sin\theta = y_\theta.$$

Remark (Debt flag). This definition extends sine and cosine beyond acute angles. Treating θ as an arbitrary real-valued radian input requires arc length and rectifiability machinery not yet developed in this volume.

Remark (Dependencies).

- Circle
- Cartesian Product
- Well-definedness of sine and cosine



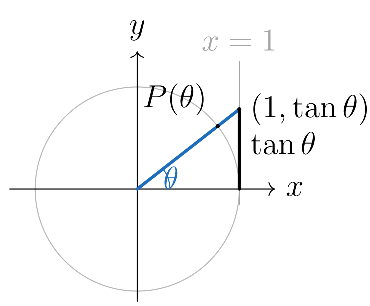
Theorem

Theorem 2.0.14 (Unit-circle agreement with right-triangle trigonometry)**.** *For $0 < \theta < 90°$, the unit-circle definitions of sine and cosine agree with the right-triangle definitions.*

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Acute-angle sine
- Acute-angle cosine
- Unit-circle cosine
- Unit-circle sine

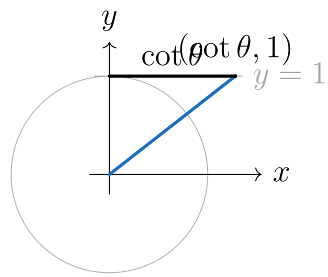


Definition

Definition 2.0.15 (Extended tangent)**.** Where $\cos\theta \neq 0$, define

$$\tan\theta = \frac{\sin\theta}{\cos\theta}.$$

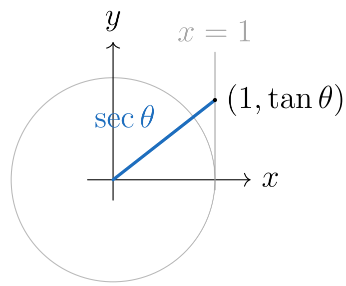
Remark (Geometric reading)*.* The tangent of θ is the signed height where the terminal ray meets the vertical tangent line $x = 1$.



Definition

Definition 2.0.16 (Extended cotangent)**.** Where $\sin\theta \neq 0$, define

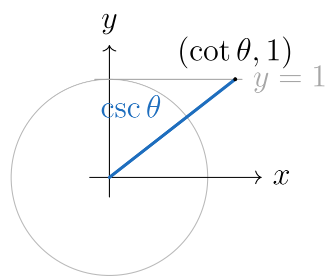
$$\cot\theta = \frac{\cos\theta}{\sin\theta}.$$



Definition

Definition 2.0.17 (Extended secant)**.** Where $\cos\theta \neq 0$, define

$$\sec\theta = \frac{1}{\cos\theta}.$$



Definition

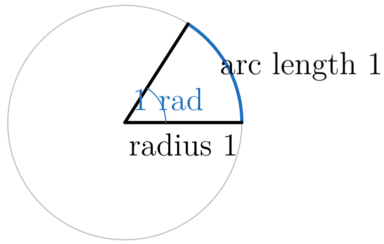
Definition 2.0.18 (Extended cosecant)**.** Where $\sin\theta \neq 0$, define

$$\csc\theta = \frac{1}{\sin\theta}.$$

Remark (Dependencies).

- Unit-circle cosine

- Unit-circle sine
- Unit-circle agreement with right-triangle trigonometry



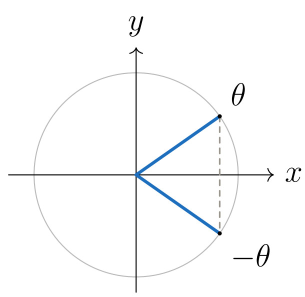
Definition

Definition 2.0.19 (Radian measure)**.** One radian is the angle that cuts off arc length 1 on the unit circle. A full turn has measure 2π radians.

Remark (Debt flag)*.* This definition uses arc length intuitively. Its rigorous foundation belongs after rectifiable curves and the construction of π from circumference.

Remark (Dependencies)*.*

- Unit-circle cosine
- Unit-circle sine
- Circle



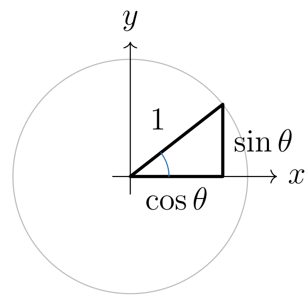
Theorem

Theorem 2.0.20 (Symmetry and periodicity of sine and cosine)**.** *For every angle in the unit-circle domain,*

$$\cos(-\theta) = \cos\theta, \qquad \sin(-\theta) = -\sin\theta,$$

and one full turn returns the same sine and cosine values.

Remark (Proof)*.* Go to proof.



Theorem

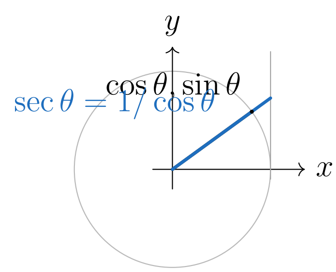
Theorem 2.0.21 (Pythagorean identity)**.** *For every angle in the unit-circle domain,*

$$\cos^2\theta + \sin^2\theta = 1.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Euclid I.47
- Unit-circle cosine
- Unit-circle sine



Theorem

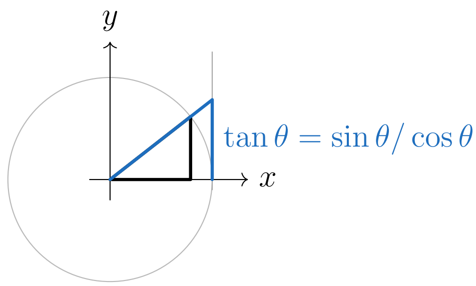
Theorem 2.0.22 (Reciprocal identities)**.** *Where both sides are defined,*

$$\sec\theta = \frac{1}{\cos\theta}, \qquad \csc\theta = \frac{1}{\sin\theta}, \qquad \cot\theta = \frac{1}{\tan\theta}.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Extended tangent
- Extended cotangent
- Extended secant
- Extended cosecant



Theorem

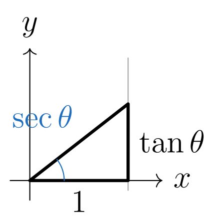
Theorem 2.0.23 (Quotient identities)**.** *Where both sides are defined,*

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Extended tangent
- Extended cotangent



Theorem

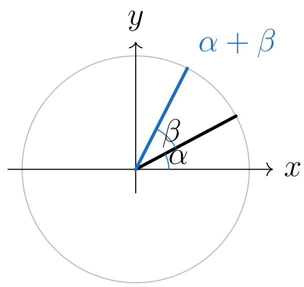
Theorem 2.0.24 (Tangent and secant Pythagorean identities)**.** *Where both sides are defined,*

$$1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Pythagorean identity
- Quotient identities
- Reciprocal identities



Theorem

Theorem 2.0.25 (Angle-sum identities)**.** *For angles in the unit-circle domain,*

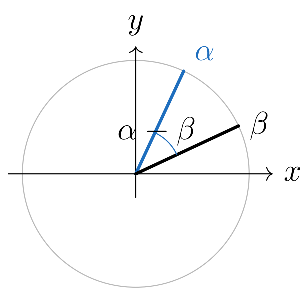
$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Pythagorean identity



Theorem

Theorem 2.0.26 (Angle-difference identities)**.** *For angles in the unit-circle domain,*

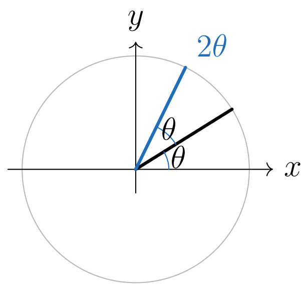
$$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta,$$

$$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Angle-sum identities
- Symmetry and periodicity of sine and cosine



Theorem

Theorem 2.0.27 (Double-angle identities)**.** *For angles in the unit-circle domain,*

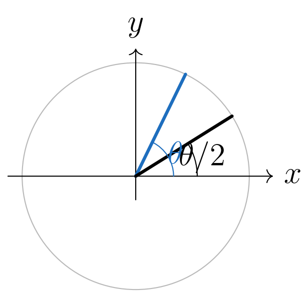
$$\sin(2\theta) = 2\sin\theta\cos\theta,$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Angle-sum identities
- Pythagorean identity



Theorem

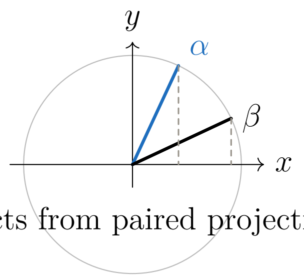
Theorem 2.0.28 (Half-angle identities)**.** *Where the signs are chosen according to the quadrant of $\theta/2$,*

$$\sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}, \qquad \cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Double-angle identities



products from paired projections

Theorem

Theorem 2.0.29 (Product-to-sum identities)**.** *For angles in the unit-circle domain,*

$$\sin\alpha\cos\beta = \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2},$$

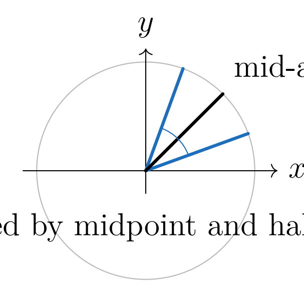
$$\cos\alpha\cos\beta = \frac{\cos(\alpha+\beta)+\cos(\alpha-\beta)}{2},$$

$$\sin\alpha\sin\beta = \frac{\cos(\alpha-\beta)-\cos(\alpha+\beta)}{2}.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Angle-sum identities
- Angle-difference identities



sums resolved by midpoint and half-difference

Theorem

Theorem 2.0.30 (Sum-to-product identities)**.** *For angles in the unit-circle domain,*

$$\sin u + \sin v = 2\sin\frac{u+v}{2}\cos\frac{u-v}{2},$$

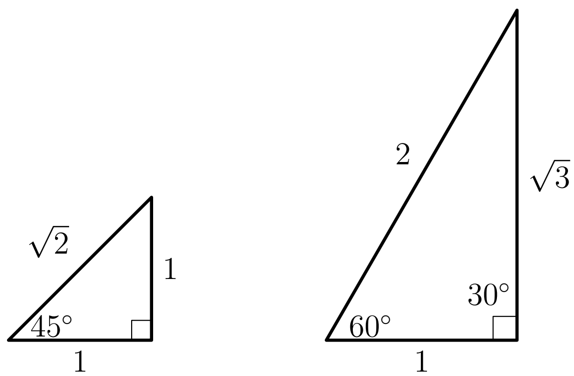
$$\cos u + \cos v = 2\cos\frac{u+v}{2}\cos\frac{u-v}{2},$$

$$\cos u - \cos v = -2\sin\frac{u+v}{2}\sin\frac{u-v}{2}.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies).

- Product-to-sum identities



Remark (Special angles). The 45° and 30°–60° values come from the isosceles right triangle and the bisected equilateral triangle. The radian entries use the radian convention above and inherit its arc-length debt.

Chapter 3

Analytical Geometry

3.1 Analytical Geometry

Remark (Status). Notes, proofs, and exercises will appear here in a future revision.

3.2 Lines

Definition

Definition 3.2.1 (Point-slope form). Let $P_1 = (x_1, y_1)$ be a point in the Euclidean plane and let $m \in \mathbb{R}$. The point-slope form of the line through P_1 with slope m is

$$y - y_1 = m(x - x_1).$$

Remark (Standard quantified statement). For every point $P_1 = (x_1, y_1) \in \mathbb{R}^2$ and every $m \in \mathbb{R}$, the corresponding line is the set of all $(x, y) \in \mathbb{R}^2$ satisfying $y - y_1 = m(x - x_1)$.

Remark (Derivation). The formula records the condition that the slope from P_1 to an arbitrary point (x, y) on the line is constant and equal to m.

Remark (Dependencies).

- Real Numbers
- Cartesian Product
- Subtraction on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition

Definition 3.2.2 (Two-point form)**.** Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be distinct points with $x_1 \neq x_2$. The two-point form of the line through P_1 and P_2 is

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1).$$

Remark (Standard quantified statement)*.* For all distinct points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ in $\mathbb{R}^2$ with $x_1 \neq x_2$, the line through P_1 and P_2 is the set of all $(x, y) \in \mathbb{R}^2$ satisfying the displayed two-point equation.

Remark (Dependencies)*.*

- Point-Slope Form
- Subtraction on $\mathbb{R}$
- $\mathbb{R}$ Is a Field

Definition

Definition 3.2.3 (General linear form)**.** The general linear form of a line in $\mathbb{R}^2$ is

$$Ax + By + C = 0,$$

where $A, B, C \in \mathbb{R}$ and $A^2 + B^2 \neq 0$.

Remark (Standard quantified statement)*.* For all $A, B, C \in \mathbb{R}$ with $A^2 + B^2 \neq 0$, the equation $Ax + By + C = 0$ determines a nondegenerate line in $\mathbb{R}^2$.

Remark (Interpretation)*.* The vector (A, B) is normal to the line. This form includes vertical lines and therefore avoids the exceptional cases present in slope-based forms.

Remark (Dependencies)*.*

- Real Numbers
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$
- $\mathbb{R}$ Is a Field

Definition

Definition 3.2.4 (Vector parametric form)**.** A line in $\mathbb{R}^2$ may be represented as the image of $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ given by

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v},$$

where $\mathbf{r}_0 \in \mathbb{R}^2$ is a point on the line and $\mathbf{v} \in \mathbb{R}^2$ is a nonzero direction vector.

Remark (Standard quantified statement). For every $\mathbf{r}_0 \in \mathbb{R}^2$ and every nonzero $\mathbf{v} \in \mathbb{R}^2$, the set $\{\mathbf{r}_0 + t\mathbf{v} : t \in \mathbb{R}\}$ is a line in $\mathbb{R}^2$.

Remark (Dependencies).

- Real Numbers
- Cartesian Product
- Function
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition

Definition 3.2.5 (Intercept form)**.** If a line meets the x-axis at $(a, 0)$ and the y-axis at $(0, b)$ with $a, b \neq 0$, its intercept form is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

Remark (Standard quantified statement). For all nonzero $a, b \in \mathbb{R}$, the line with intercepts $(a, 0)$ and $(0, b)$ is the set of all $(x, y) \in \mathbb{R}^2$ satisfying $x/a + y/b = 1$.

Remark (Derivation). The intercept form follows by applying the two-point form to the points $(a, 0)$ and $(0, b)$ and then rearranging the resulting equation.

Remark (Dependencies).

- Two-Point Form
- $\mathbb{R}$ Is a Field

Definition

Definition 3.2.6 (Affine parametric line in $\mathbb{R}^n$)**.** Given distinct points $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the line through them is

$$\mathbf{x}(t) = \mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1), \qquad t \in \mathbb{R}.$$

Remark (Standard quantified statement). For all distinct $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the set $\{\mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1) : t \in \mathbb{R}\}$ is the affine line through $\mathbf{x}_1$ and $\mathbf{x}_2$.

Remark (Interpretation). For $0 \leq t \leq 1$, the same formula describes the line segment joining $\mathbf{x}_1$ and $\mathbf{x}_2$. For arbitrary real t, it describes the entire affine line.

Remark (Dependencies).

- Real Numbers
- Cartesian Product

- Subtraction on $\mathbb{R}$
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition

Definition 3.2.7 (Normal form)**.** A line in $\mathbb{R}^2$ may be written in normal form as

$$x \cos \alpha + y \sin \alpha - p = 0,$$

where $p \geq 0$ is the distance from the origin to the line.

Remark (Standard quantified statement)*.* For every $p \geq 0$ and every real angle α, the equation $x \cos \alpha + y \sin \alpha - p = 0$ describes a line whose unit normal direction is $(\cos \alpha, \sin \alpha)$.

Remark (Dependencies)*.*

- General Linear Form
- Acute-angle sine
- Acute-angle cosine
- Order on $\mathbb{R}$

Definition

Definition 3.2.8 (Symmetric form)**.** Given a point (x_1, y_1) and direction cosines $(\cos \theta, \sin \theta)$, the symmetric form of a line is

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r.$$

Remark (Standard quantified statement)*.* For every point (x_1, y_1) and every direction with nonzero displayed denominators, the symmetric equation describes the same line as the parametric equations $x = x_1 + r \cos \theta$ and $y = y_1 + r \sin \theta$.

Remark (Dependencies)*.*

- Vector Parametric Form
- Acute-angle sine
- Acute-angle cosine
- $\mathbb{R}$ Is a Field

Bibliography

Euclid. *The Thirteen Books of Euclid's Elements*. Dover Publications, 1956.